

Primitive-Ring Symmetry Index Specification

Stage 7R: Compact Catalog-Bound Derived Ring Action

mdstats

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Purpose and boundary

`primitive_ring_symmetry.py` stores the exact action of one `PeriodicNetSymmetry` on one `PrimitiveRingCatalog`. It is a derived index, not a field of the core net-symmetry result.

Runtime/API target:

mdstats 0.19.19a0

The ownership chain is

```
PeriodicNetView
  -> PeriodicNetSymmetry
PrimitiveRingCatalog
  -> PrimitiveRingIndex
      + PeriodicNetSymmetry
      -> PrimitiveRingSymmetryIndex
```

This split prevents a net automorphism group from depending on a particular ring search bound and binds every ring orbit explicitly to the exact catalog that supplied it.

Motivation

A periodic-net automorphism group is defined by the decorated graph view. A primitive-ring action additionally depends on:

- which ring family was enumerated;
- the minimum and maximum search sizes;
- resource-completeness metadata; and
- the exact ring-catalog digest.

Persisting ring data inside `PeriodicNetSymmetry` merged two scientific sources, repeated complete `PrimitiveRingKey` payloads for every operation, and allowed a ring action to be restored without recording the exact catalog provenance.

Mathematical action

For normalized operation

$$g(i, \mathbf{n}) = (\pi_g(i), A_g \mathbf{n} + \boldsymbol{\tau}_i^g),$$

and canonical zero-shift ring representative R_k , the derived action records

$$g(R_k) = T_{s_{gk}} R_{p_g(k)},$$

plus one cycle parameterization

$$(c_{gk}, \epsilon_{gk}), \quad \epsilon_{gk} \in \{-1, +1\}.$$

$p_g(k)$ is stored as an integer ring position, not as a repeated full key.

Normalized representatives compose with cocycle

$$\hat{g}\hat{h} = T_{c(g,h)}\hat{g}\hat{h}.$$

Therefore exact placement composition requires

$$s_{gh,k} = A_g s_{h,k} + s_{g,p_h(k)} - c(g, h).$$

The builder verifies this identity for every operation pair and every stored ring. It also verifies composition of cycle orientation and start position.

Public API

```
@dataclass(frozen=True, slots=True)
class RingSymmetryImage:
    target_ring_position: int
    target_image_shift: LatticeShift
    parameterization: CycleParameterization

@dataclass(frozen=True, slots=True)
class PrimitiveRingSymmetryIndex:
    periodic_net_symmetry_digest: str
    periodic_net_view_digest: str
    topology_graph_digest: str
    primitive_ring_catalog_digest: str
    complete_for_ring_sizes_up_to: int
    source_search_completed_without_resource_truncation: bool
    ring_keys: tuple[PrimitiveRingKey, ...]
    action_table: tuple[tuple[RingSymmetryImage, ...], ...]
    ring_orbits: tuple[tuple[int, ...], ...]
    ring_stabilizers: tuple[tuple[int, ...], ...]
    digest: str
```

Builder:

```
build_primitive_ring_symmetry_index(
    view: PeriodicNetView,
    symmetry: PeriodicNetSymmetry,
    ring_index: PrimitiveRingIndex,
    *,
    max_composition_checks: int = 5_000_000,
) -> PrimitiveRingSymmetryIndex
```

Convenience methods include:

```
ring_position(ring_key)
ring_key(position)
ring_image(operation_index, ring_key)
target_ring_key(image)
map_placement(symmetry, operation_index, placement)
```

Source invariants

Construction requires:

1. `symmetry.periodic_net_view_digest == view.digest`;
2. matching topology graph digests;
3. the same framework edge-orbit set in the view and ring catalog;
4. one exact action row per symmetry operation;
5. one exact image per catalog ring; and
6. a complete group-action homomorphism under the stored translation cocycle.

The index stores the catalog completeness bound but does not reinterpret it. Ring orbits are complete only for the supplied catalog.

Compact persistence

Schema:

`mdstats.primitive-ring-symmetry.v1`

Complete ring keys appear once. Table cells carry only:

```
target ring position
three-integer image shift
start position
orientation
```

For the Na-LTA gate, this reduces the ring-action JSON from repeated full-key records to approximately 1 MB while preserving all 96-by-82 exact images.

Serialization

`from_dict()` requires all three source objects:

```
PrimitiveRingSymmetryIndex.from_dict(
    payload,
    view=view,
    symmetry=symmetry,
    ring_index=ring_index,
)
```

The method rebuilds and revalidates the complete action table. A payload cannot be transferred to another symmetry group or another primitive-ring catalog even when the topology graph is the same.

Edge cases

- empty ring catalog: supported as an empty finite domain only if the action table shape is consistent;
- ring catalog with different edge set: rejected;
- different view policy on the same topology: rejected by view digest;
- catalog with same keys but different provenance: rejected by catalog digest;
- nonsymmorphic operation: image shifts are composed with the exact cocycle;
- translated input placement: mapped by $A_g \mathbf{t} + \mathbf{s}_{gk}$;
- resource limit exceeded during homomorphism validation: no partial index is returned.

Focused tests

- identity, rotation, and reflection actions;
- ring orbit and stabilizer partitions;
- parallel-edge and occurrence orientation handling;
- source-bound serialization;
- wrong-group and wrong-catalog rejection;
- exact translated-placement action; and
- all 82 Na-LTA rings under the 96-operation group.

References

- [1] S. J. Chung, Th. Hahn, and W. E. Klee, *Acta Cryst. A* **40**, 42-50 (1984), doi:10.1107/S0108767384000088.
- [2] O. Delgado-Friedrichs and M. O’Keeffe, *Acta Cryst. A* **59**, 351-360 (2003), doi:10.1107/S0108767303012017.